Lightweight TRNG Based on Multiphase Timing of Bistables

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***Abstract*—The paper presents a concept of a True Random Number Generator (TRNG) that utilizes phase noise of a pair of ring oscillators (ROs) to increase the variance of the initial condition of a bistable. For this purpose a special TRNG D-latch architecture (TDL) has been proposed, which can either operate in the oscillatory ring-oscillator mode or the nearly-metastable mode. The RO mode increases the probability of the nearly metastable operation of the TDL, which in turn increases the mean value and variance of the resolve time. Moreover, due to the oscillatory nature of the TDL metastability, the resolve time can be easily measured and used for further randomness harvesting in the TRNG. The proposed TRNG uses a pair of TDLs to reduce sensitivity to process or temperature variation, whereas TDLs' individual resolve time, their resolve time difference and logical state contribute to randomness. In the article, the impact of the devices' imbalance (tolerance) resulting from process variation on the resultant entropy is also explored. The proposed TRNG based on TDLs is scalable and has been implemented in modern CPLD and FPGA devices whereas the bit rate reached up to 1 Mbit. The article also discusses theoretical issues related to a transformation of phase and resolve time probability density functions and their influence on TRNG parameters.**

***Index Terms*—CPLD, FPGA, metastability, randomness, TRNG.**

I. Introduction

RANDOM number generators (RNGs) are indispensable part of cryptographic protocols, therefore they are often used in modern electronic and telecommunication systems. Due to popularity of digital communication, it is necessary to use encryption techniques, which in turn require random number sequences. In some applications these sequences cannot be a result of deterministic algorithms and have to meet strict statistical tests requirements. The devices which meet such requirements usually harvest entropy from physical phenomena like phase noise (jitter), instability of bistables (metastability), and randomness of various electrical parameters. In order to distinguish RNGs with the physical sources of entropy from those without such sources, these RNGs are called true random number generators (TRNGs).

Manuscript received November 6, 2015; revised February 29, 2016; accepted March 29, 2016. Date of publication June 16, 2016; date of current version July 22, 2016. This paper was recommended by Associate Editor T. S. Gotarredona.

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Digital Object Identifier 10.1109/TCSI.2016.2555248

There are several modern TRNG solutions that benefit from the flip-flop metastability or jitter resulting from the ring-oscillator (RO) phase noise. A major disadvantage of the metastability based solutions is the necessity of operation near an unstable point [1]. In theory, reaching the neighborhood of this point is necessary to observe the influence of noise processes on a response (state) of a bistable (flip-flop or D-latch). Due to the small diameter of the metastable point neighborhood on the state plane, any inaccuracy (resulting from transistor tolerances or different flip-flop path delays) affecting the symmetry of the bistable can cause its unwanted deterministic behavior [2], [3]. There are some modern TRNG solutions utilizing metastability, in which authors try to solve this issue. Hata and Ichikawa [4] proposed the solution implemented in an FPGA, which is composed of look-up tables (LUTs) forming RS flip-flops with low tolerances of set (S) and reset (R) propagation delays. This way the authors obtained flip-flops with almost identical parameters, whereas the metastability occurs when the R and S lines get both active signals simultaneously. The discussed solution does not require specialized adaptation mechanisms for reaching the metastability, nevertheless high flip-flop redundancy is recommended.

Another solution for the metastability-based TRNG is to employ a precise programmable delay line (PDL) or a digital clock manager (DCM) device, which adapts the violation of the set-up and hold time of a single D flip-flop. The proper adjustment of the clock to data lead-in time causes the unpredictability of the output state. Majzoobi *et al.* [1] presented a metastability-based TRNG with a proportional-integral controller of the metastability proximity. In this case, the initial condition of a bistable (D flip-flop) is controlled by variable lead-in time of the clock and data signal active slopes. Due to the use of adaptive control, the device provides robustness against attacks and varying process, voltage or temperature parameters (PVT).

An interesting way to control the metastability proximity is to directly adjust an initial input voltage of inverters forming a bistable. Such a solution has been proposed in Tokunaga *et al.* [5]. In this approach, depending on the flip-flop time of response, a specially designed subcircuit adapts the initial voltage on the bistable's input. The main disadvantage of this approach is the necessity of a silicon level implementation.

Barangi *et al.* [6] presented a recent research, which showed that the metastable state observed in the straintronics magnetic tunneling junction (MTJ) can be used in a TRNG. The principle of this approach is based on the fact that the MTJ settles into the metastable point when a physical stress is present. The removal of the stress causes the MTJ relaxation and switching between

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high and low resistances. According to the results presented in the article the straintronics MTJ TRNG successfully passes all of the NIST tests and can operate with frequencies up to 510 MHz.

Many TRNGs utilize the phase randomness of ROs, or jitter of analog voltage controlled oscillators in phase-locked loops (PLLs) [7]. In these types of TRNGs, the randomness harvesting and its control is easier than in the metastability based TRNGs. Nevertheless, RO and PLL based TRNGs are not without drawbacks. There are many effects, especially in the RO-based TRNGs, which hinder their usage, like the frequency locking, aging, technology dependence or high power consumption. Moreover, there are some papers showing easy methods for frequency attacking [8]; however, in the case of the metastability based solutions the frequency attacks require knowledge of the bistable's timing [9]. Rahman *et al.* [10] proposed some additional solutions for a RO-based TRNG to improve randomness across time and during attacks, like the extra power supply noise and self-compensation mechanism. Böhl *et al.* [11] proposed the RO-based TRNG with the (so-called) on-line testability based on a single oscillator with multiple taps. The discussed concept passes all standard statistical tests and reaches 56.3 kbit/s of the bit rate and can be implemented in an FPGA device.

An interesting enhancement to the TRNGs based on bistables was proposed by Varchola and Drutarovsky [12]. In their approach named TERO (transition effect ring oscillator) the oscillatory metastability is used as a randomness source. The TERO's topology is similar to an RS flip-flop consisting of NAND gates, in which both the reset and set inputs are connected together [13]. In theory, the positive edge on the common node of R and S (called the "control" line in Varchola *et al.* [14]) causes the metastable behavior, because the lead-in interval (δ) of R and S signal edges is equal to zero [15]. Usually the resolve time (t_r) of the flip-flop forced to the metastable operation is random and relatively short, though, its average depends on the active elements time constants resulting from the technology limitations [16]. However, Varchola *et al.* proposed additional modification of an ordinary RS flip-flop, in which the propagation delay (τ) of active elements is intentionally increased. Such a modification leads to an oscillatory response of the bistables autonomous phase, if $\tau > RC$, where R is the output resistance and C is the input capacitance in a small signal model of logical gates [15]–[17]. Therefore, despite the relatively fast circuits used for the TERO implementations (Actel FPGA), the obtained t_r is of the range of hundreds of ns, whereas the TERO response is oscillatory. This makes the t_r randomness harvesting easy, since the number of oscillations can be used as a t_r estimate. The TERO provides 250 kbit/s throughput and a lightweight implementation. Nevertheless, it might be sensitive to the delay path inequalities, especially in the control line, which must ensure identical delays of gates inputs to maintain the metastable operation. In this solution the metastability proximity depends on both, the initial noise amplitude and gates (control line delays) matching.

This paper presents both the novel TRNG D-latch (TDL) and its implementation in a TRNG. Author proposes to use ROs to adjust entropy of the initial condition of an autonomous phase of the TDL, and to increase the probability of its metastable operation and resultant t_r randomness. For this reason an additional pre-autonomous phase (mode) of operation between bistable's clear and metastable phases was added. This is the main difference to the TERO solution, in which the entropy of the bistable's initial condition results only from the momentary internal thermal and flicker noises. The RO mode is obtained by the special design of the proposed TDL, in which the feedback loop paths can change. This way the entropy of the bistable's initial condition significantly rises and roughly depends on the preceding number of ROs oscillations. Whereas the TERO solution cannot adjust the bistable initial condition, the proposed solution can adjust it in the pre-autonomous mode. Moreover, randomness in the proposed TRNG is harvested from the parameters of two bistables' (TDLs') transient response, i.e., their t_r difference of oscillatory responses, parity of metastable oscillation number, and logical states. Additionally, the symmetry of the proposed design increases its robustness and resistance to temperature and supply voltage variations.

The remainder of this paper is organized as follows. In Section II, the principle of multiphase timing of the TDL and its modes are described. Section II-A focuses on the description of the phase noise $(\widetilde{\varphi})$ of TDL's ROs and corresponding voltage random variables $\{\widetilde{U}_Q, \widetilde{U}_{\overline{Q}}\}$, used for an enhancement of the metastability. Section II-B describes the transformation of random variables $\{\widetilde{U}_Q, \widetilde{U}_{\overline{Q}}\}$, to the TDL's random resolve time variable $(\widetilde{T}_r)$, in the metastable mode. Finally, Section II-C and D show the relationship between $\widetilde{T}_r$ and TDL circuit parameters, and environmental conditions. The implementation of the TRNG based on TDLs, methods of $\widetilde{T}_r$ randomness harvesting, measurements, and statistical testing results are presented in Section III.

II. TRNG Concept and Mathematical Modeling

The TRNG device described in this paper is based on the unpredictability of a metastability process in the structures topologically similar to D-latches (TDLs), which are encapsulated in reprogrammable devices. In general, every bistable operating near its unstable solution is very susceptible to the noise processes affecting its initial condition in a pre-autonomous phase of operation [when a positive feedback loop is inactive, see Fig. 1(a)]. In this phase the noise influences both the final solution (logical state) and t_r [defined in Fig. 1(a)]. Whereas the autonomous phase of operation is understood as the interval, in which the positive feedback loop is active. In bistables, a diameter of the neighborhood of the metastable point $\{U_{QM}, U_{\overline{Q}M}\}$ leading to the unpredictability of t_r, is very small. This results from the large sensitivity of t_r to the difference of U_Q and $U_{\overline{Q}}$, and a relatively small amplitude of the noise $N_1(t)$ and $N_2(t)$. Therefore, the author proposes to increase the amplitude of the noise additive to U_Q and $U_{\overline{Q}}$, in order to increase the resultant variance of the resolve time (the resolve time understood as a random variable will be denoted as $\widetilde{T}_r$). For this reason, the phase noise of two ROs is applied to U_Q and $U_{\overline{Q}}$ in the pre-autonomous operation of the bistable. The TDL concept utilizing this idea [shown in

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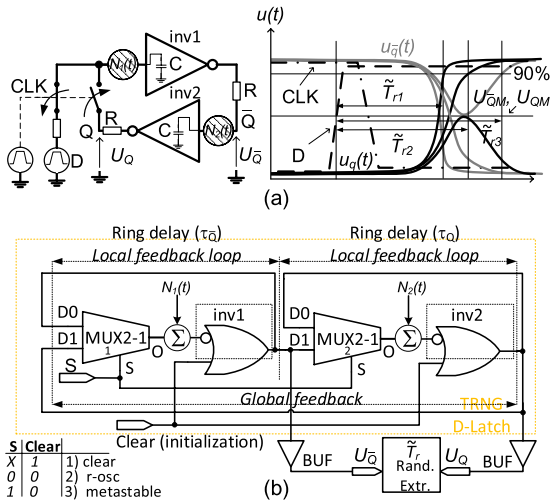

Fig. 1. Bistables with two inverters: (a) a popular electric model and its metastable behavior in the time domain. Random resolve time can be observed $(\widetilde{T}_{r1}, \widetilde{T}_{r2}, \widetilde{T}_{r3})$ for a constant clock and data lead-in time, due to the $N_{1,2}(t)$ noise sources; (b) the proposed TDL topology.

Fig. 1(b)] is built of two symmetric inverters (inv$_1$ and inv$_2$) that can operate in three different configurations (modes), which are: i) independently disconnected (cleared), ii) switched to the ring-oscillator mode (r-osc) for a predefined t_{rosc} interval; and finally iii), switched back to the autonomous (metastable) operation with random $\widetilde{U}_Q = \widetilde{u}_q(t = t_{\mathrm{rosc}})$ and $\widetilde{U}_{\overline{Q}} = \widetilde{u}_{\bar{q}}(t = t_{\mathrm{rosc}})$, resulting from the random phases $\widetilde{\varphi}_{\overline{Q}}$ and $\widetilde{\varphi}_Q$ of inv$_1$ and inv$_2$ respectively. This capability has been called the multiphase timing of bistables. The last mode of the TDL is used for the randomness harvesting based on the $\widetilde{T}_r$ random variable [$\widetilde{T}_r$ *Rand. Extr.* in Fig. 1(b)]. The harvesting can be performed either by $\widetilde{T}_r$ measurement or its comparison with $\widetilde{T}_r$ of another TDL device. Therefore, this concept requires a sequence of consecutive 1–3 modes in order to generate a random variable. A complete TRNG solution (discussed in Section III) utilizes the $\widetilde{T}_r$ variability of two identical TDLs (A and B respectively). The resolve time of devices ($\widetilde{T}_{rA}$ and $\widetilde{T}_{rB}$) is compared by an arbiter [15]. The further theoretical considerations focus on the mathematical analysis of a single TDL.

A. Phase Description During TDL r-osc Mode

The propagation delays of inv$_1$, inv$_2$ and multiplexers (τ_Q and $\tau_{\overline{Q}}$), operating with a local feedback loop, cause the oscillatory behavior of both rings in the r-osc mode [mode 2, when $S = 0$ in Fig. 1(b)]. In this mode, the thermal and flicker noise cause the momentary variability of oscillatory $u_q(t)$ and $u_{\bar{q}}(t)$ voltages, which in turn alter the $\widetilde{\varphi}_Q(t)$ and $\widetilde{\varphi}_{\overline{Q}}(t)$ phases, when $t \in (0, t_{\mathrm{rosc}})$. The circuit in Fig. 2 represents the model of either inv$_1$ or inv$_2$ and corresponding $\widetilde{\varphi}_Q(t)$ and $\widetilde{\varphi}_{\overline{Q}}(t)$ phases during the r-osc mode of the TDL. The model can be described by a stochastic delay differential equation (SDDE) (1), where $\widetilde{U}_C$ denotes a random variable representing nondeterministic behavior on the inv$_1$ or inv$_2$ input, $f(\cdot)$ is a hyperbolic tangent transfer function of inv$_1$ or inv$_2$ [18], τ is the propagation delay, C is the input capacitance, R the average output resistance of

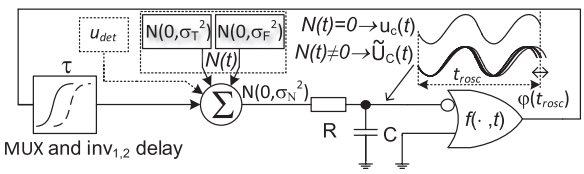

Fig. 2. Model of the TDL oscillators in the r-osc mode.

each inverter, and $N(t)$ is the stochastic process representing the voltage noise [$N_1(t)$ or $N_2(t)$ in Fig. 1(b)]

$$\dot{\widetilde{U}}_C = \frac{d\widetilde{U}_C}{dt} = \frac{f(\widetilde{U}_C, t-\tau) - \widetilde{U}_C + N(t)}{RC}. \tag{1}$$

The $N(t)$ component represents both the thermal noise $(\mathcal{N}(0, \sigma_T^2))$ and the flicker noise $(\mathcal{N}(0, \sigma_F^2))$, where the influence of the deterministic components of the power supply or the frequency injection attack is denoted as u_{det}. The influence of u_{det} is discussed in details in Section II-D. In general, if $u_{\mathrm{det}} = 0$, the $N(t)$ probability distribution function (pdf) results from the sum of two Gaussian variables and is Gaussian too, with the $\sigma_N^2 = \sigma_T^2 + \sigma_F^2$ variance, and zero average $(\mu_N = 0)$. It is worth mentioning that the flicker noise is of 1/f type and some researchers claim that its presence causes the correlation of the phase walk $(\widetilde{\varphi}(t))$ in several subsequent RO periods [19]. However, in the TDL a long $\widetilde{\varphi}_Q(t)$ and $\widetilde{\varphi}_{\overline{Q}}(t)$ history contributes to the randomness and avoids the correlation of final random walks of inv$_1$ and inv$_2$. If $N(\cdot) = 0$ in (1), then the voltage on the inverter's input becomes deterministic (denoted as $u_c(t)$) and can be derived as a solution of the first order differential delay equation (DDE) (2). In order to find u_c, $f(\cdot)$ is simplified to the linear form, in which g represents the dynamic slope of the inverter $f(\cdot)$ transfer function

$$\frac{du_c}{dt} = \frac{-gu_c(t-\tau) - u_c}{RC}. \tag{2}$$

Applying the Laplace transform to (2) one can obtain (3) in the s-domain

$$sU_C(s) - u_c(0) = -\frac{U_C(s)}{RC} - \frac{g}{RC}e^{-\tau s}U_C(s). \tag{3}$$

Assuming that $u_c(0) = 0$, (3) leads to (4)

$$s + \frac{1 + ge^{-\tau s}}{RC} = 0. \tag{4}$$

Equation (4) can be solved by finding zeros and represented as (5) in the time-domain

$$u_c = \sum_{k=-\infty}^{\infty} c_k e^{\left[\frac{1}{\tau}W_k\left(-\frac{\tau g}{RC}e^{\frac{\tau}{RC}}\right) - \frac{1}{RC}\right]t} \tag{5}$$

where $W_k(\cdot)$ is a k-th branch of the Lambert function, which is complex-valued and c_k is a coefficient determined from the initial condition and history function of (2) [20]. Therefore, according to (5) u_c is periodic and can be represented as a sum of Fourier series; however, it is limited to lowest frequency components [17]. The (5) explains the deterministic oscillatory behavior of TDL's r-osc mode and is useful in calculation of

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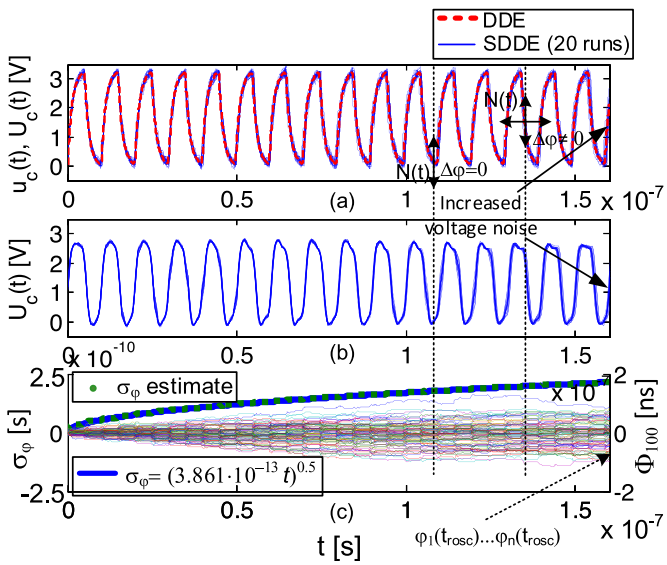

Fig. 3. Behavior of the TDL in the r-osc mode: (a) modeled with DDE (2) and SDDE (1), where 20 runs are shown; (b) measured time domain behavior of the Cool-RunnerII circuit with the MDO3024 oscilloscope; (c) the phase walk for 100 realizations— Φ_{100}, and the phase $\sigma_\varphi(t_{\rm rosc})$ standard deviation.

the impulse sensitivity function (ISF) of ROs to the noise when $N(\cdot) \neq 0$ [21], [22]. A quite intuitive approach to a phase noise identification was proposed by Hajimiri [21], [22] and assumes that a random phase walk realization ($\widetilde{\varphi}$) is the result of an infinite sum of momentary phase fluctuations ($\Delta\varphi$) caused by changes of $\widetilde{U}_C$ resulting from the momentary values of $N(t)$, i.e., $\widetilde{\varphi}(t_{\rm rosc}) = \int_0^{t_{\rm rosc}} \Delta\varphi dt$, in $t_{\rm rosc}$ time. Therefore, the resultant $\widetilde{\varphi}$ phase can be calculated by the convolution (6) of the $N(t)$ stochastic process realization with the periodic ISF denoted as $\Gamma(\omega t)$

$$\widetilde{\varphi}(t_{\rm rosc}) = \int_0^{t_{\rm rosc}} \Gamma(\omega t) N(t) dt. \tag{6}$$

The ISF represents generator's (momentary) sensitivity to the noise realization in time. In short, the ISF reaches highest value when $|du_c(t)/dt|$ reaches highest value too, because under such condition the $\widetilde{U}_C$ voltage variations cause highest phase changes. In the opposite case, when $du_c(t)/dt = 0$, $N(t)$ does not affect the generator's phase but $\widetilde{U}_C$ [$\widetilde{U}_Q$ or $\widetilde{U}_{\overline{Q}}$ in Fig. 1(b)] only. Assuming a single harmonic ISF approximation, it can be represented as $\Gamma(2\pi f_C t) \propto (1/2A\pi f_C)\cos(2\pi f_C t)$, where A is an oscillation amplitude and f_C an oscillation frequency of a RO. In Fig. 3(a) one can see a periodic increase of the phase noise of the $\widetilde{U}_C$ ($\widetilde{U}_Q$ or $\widetilde{U}_{\overline{Q}}$) voltage in time. The results were obtained with (1) and the Euler-Maruyama method [23]. The total phase noise after $t = t_{\rm rosc}$ is a collection of the $\widetilde{\varphi}_n(t_{\rm rosc})$ realizations, i.e., $\Phi_n = (\widetilde{\varphi}_1(t_{\rm rosc}), \widetilde{\varphi}_2(t_{\rm rosc}), \ldots, \widetilde{\varphi}_n(t_{\rm rosc}))$, and results from n consecutive runs of the r-osc mode. The estimate of the $N(t)$ model's σ_N standard deviation can be done by the measurement of the phase variance ($\sigma^2_\varphi(t_{\rm rosc})$) in a physical circuit and a numerical phase walk calculation with either (1) or (6). This way, when the variance definition (7) is applied,

TABLE I
ROs FREQUENCIES AND STANDARD DEVIATIONS FOR VARIOUS IMPLEMENTATIONS AND OPERATING CONDITIONS

Device	Conditions	$f_C[MHz]$	$t_{rosc}[ns]$	$\sigma_\varphi[ps]$	$\sigma_N[mV]$
FPGA	27^oC	583.5	438	154	37
	7^oC	594.0	431	110	27
	$0.9V_{core}$	501.8	510	134	25
CPLD	27^oC	108.7	294	337	49
	7^oC	111.5	287	254	38
	$0.9V_{core}$	91.08	351	321	36

the σ_N of the model can be iteratively adjusted to the behavior (jitter) of a real circuit

$$\sigma^2_\varphi(t_{\rm rosc}) = E\left[(\widetilde{\varphi}(t_{\rm rosc}) - \mu(t_{\rm rosc}))^2\right] \tag{7}$$

where the $\widetilde{\varphi}(t_{\rm rosc})$ phase walk and $\mu(t_{\rm rosc})$ average after $t_{\rm rosc}$ are obtained from the solution of (1) or from (6). The estimated σ_N and phase walk standard deviations are listed in Table I. The first tested physical TDL implementation utilized a CPLD Cool-RunnerII circuit, in which the MUX2 $- 1_{1-2}$ and inv$_{1-2}$ interconnections had the *noreduce* flag. The behavior of the device in the time domain is presented in Fig. 3(b). The second implementation was based on an FPGA Spartan 6 device, in which the MUX2 $- 1_{1-2}$ and inv$_{1-2}$ components were substituted with LUT3-L and LUT2-L blocks respectively. This allowed for fast internal slice connections without a variable delay introduced by the autorouter. Table I also lists measured f_C frequencies and $\sigma_\varphi(t_{\rm rosc})$ variances for corresponding $t_{\rm rosc}$. The f_C frequency measurements were performed with the Agilent 53230A timer, which measured the distribution of the carry output signals of binary counters connected to inv$_{1-2}$. The counters were 32 and 256 for CPLD and FPGA devices respectively. The indirect internal measurement was necessary due to high frequencies especially in the case of FPGA devices. The experiment was conducted in 27 °C and 7 °C temperature with the normal, and 10% decreased core power supply. The inter-class f_C tolerance resulting from the process variation for 5 Cool-RunnerII devices (in Digilent's Cool-RunnerII CPLD boards) and 4 Spartan 6 devices (in Digilent's Nexys 3 boards) in ambient temperature, was less than 3.5%. However, the intra-class tolerance (of different ROs within one device) did not exceed 1%.

Due to the major influence of τ on the f_C resulting from high interconnection delays, $\tau \approx 1/2f_C$ was assumed in (1) [17]. The remaining inverter model parameters were adjusted to satisfy physical $\widetilde{T}_r$ variance ($\sigma_{\widetilde{T}_r}$) and (9)–(13) in the oscillatory metastable mode of particular devices. For this reason the description of metastable behavior is the subject of next section. The inverter parameters used in calculations were: $\tau = 4.076$ ns, $RC = 1.33$ ns, and $g = 21$, and correspond to the physical behavior of the TDL during the r-osc mode of a Cool-RunnerII device.

Fig. 3(c) shows exemplary phase walks obtained with (6) for $t_{\rm rosc} = 160$ ns. The experiments results also allowed to estimate the coefficient of the $\sigma^2_\varphi \propto t_{\rm rosc}$ proportion, i.e., $\sigma^2_\varphi(t_{\rm rosc}) = c_\varphi \cdot 10^{-13} \cdot t_{\rm rosc}$, where $c_\varphi = 3.861$ for Cool-RunnerII. Additional measurements for FPGA Spartan 6 devices (not shown in Fig. 3) led to $c_\varphi = 0.541$, $\tau = 0.679$ ns, $RC = 266$ ps, and $g = 18$. Random values of $\widetilde{\varphi}(t = t_{\rm rosc})$ obtained either numerically

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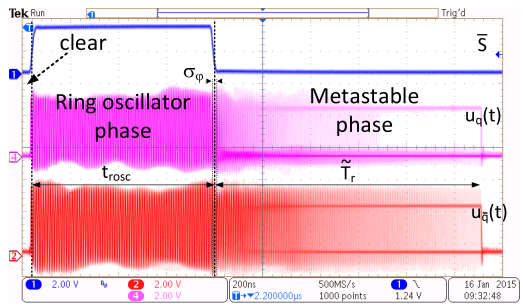

Fig. 4. Oscillatory behavior of the TDL during the metastable operation. Pre-phases are marked: clear and r-osc mode.

or experimentally are of the Gaussian type according to the Lilliefors test of 10^4 $\widetilde{\varphi}(t_{\text{rosc}})$ samples (Φ_{10^4}), at the 5% significance level. Therefore $\widetilde{\varphi}_Q(t)$ and $\widetilde{\varphi}_{\overline{Q}}(t)$ can be considered as phase walks of two Wiener processes.

B. Initial Condition Distribution and Resolve Time of TDL in Metastable Mode

The Φ phase noise in $t = t_{\text{rosc}}$ transforms into the voltage noise through the ISF, causing the increased noise of the metastable initial condition $\widetilde{U}_Q = \widetilde{u}_q(t_{\text{rosc}})$ and $\widetilde{U}_{\overline{Q}} = \widetilde{u}_{\overline{q}}(t_{\text{rosc}})$, when $S = 0 \rightarrow 1$. The high noise is necessary to provide a high variance of the $\widetilde{T}_r$ variable during the metastable mode of the TDL. In short, the deterministic relationship between the initial condition voltages $\{U_Q, U_{\overline{Q}}\}$ and the resolve time of a bistable, can be described with a logarithmic function (8) [15], [16]

$$t_r = \tau_1 \ln \frac{u_1}{|U_Q - U_{\overline{Q}}|} \tag{8}$$

where τ_1 and u_1 result from the inverter capacitances and transconductances [16]. When a denominator in (8) is treated as a random variable, i.e., $\widetilde{Z} = \widetilde{u}_Q(t = t_{\text{rosc}}) - \widetilde{u}_{\overline{Q}}(t = t_{\text{rosc}})$, the variance of $\widetilde{T_r}$ depends roughly on the bias of $\widetilde{Z}$, which in turn depends on the tolerances of inverters' parameters in (2) [2], [3]. For this reason, in opposite to other solutions like in Varchola *et al.* [13], [14], the $\widetilde{Z}$ variance in the TDL results from the preceding r-osc mode, in which the random phases $\widetilde{\varphi}_Q(t = t_{\text{rosc}})$ and $\widetilde{\varphi}_{\overline{Q}}(t = t_{\text{rosc}})$ of inv_1 and inv_2, determine the random $\{\widetilde{u}_Q(t = t_{\text{rosc}}), \widetilde{u}_{\overline{Q}}(t = t_{\text{rosc}})\}$ voltages.

Equation (8) is usually applied to bistables with a non-oscillatory response during the autonomous (and metastable) operation; however, in circuits with high τ_Q and $\tau_{\overline{Q}}$ in comparison to their RC constants, the diminishing oscillations over a time can be observed [17]. Such a behavior of the TDL occurs in physical circuits regardless of the implementation, because of the relatively high τ_Q and $\tau_{\overline{Q}}$ in comparison to rise/fall times, which are necessary for the preceding r-osc mode. Fig. 4 shows the metastable oscillatory behavior of the Cool-RunnerII implementation of the TDL during the transition from the r-osc to the metastable mode. However, the presence of the diminishing oscillations is desired, because it simplifies the $\widetilde{T}_r$ measurement. The TDL in the metastable mode can be described with the nonlinear DDE set (9)

$$\begin{cases} RC\frac{du_q}{dt} + u_q = f_1(u_{\bar{q}}, t - \tau) \\ RC\frac{du_{\bar{q}}}{dt} + u_{\bar{q}} = f_2(u_q, t - \tau) \end{cases} \tag{9}$$

where $\tau = \tau_Q = \tau_{\overline{Q}}$ and $f_1(u(t)) = f_2(u(t)) = V_s \tanh[k_u(-u(t-\tau) + (V_S/2)) + 1]$ is assumed, whereas V_s stands for the supply voltage and k_u is the slope coefficient. When the small signal model is applied to f_1 and f_2, then $f_{1,2} = -gu(t - \tau)$, where $g = k_u V_s$ stands for the small signal gain, used in (2). The small signal form of the transfer function can be applied to (9), which leads to the second order differential (10)

$$\left[\frac{RC}{g}\right]^2 \frac{d^2 u_q}{dt^2} + 2\frac{RC}{g^2}\frac{du_q}{dt} + \frac{1}{g^2}u_q - u_q(t - 2\tau) = 0. \tag{10}$$

If a solution of the exponential form is assumed [17], i.e., $u_q(t) = Ce^{\Theta t}$, where C is a constant and Θ is a complex number, (10) leads to a characteristic (11) [17]

$$\left[\frac{RC}{g}\right]^2 \Theta^2 + 2\frac{RC}{g^2}\Theta + \frac{1}{g^2} - e^{-2\Theta\tau} = 0. \tag{11}$$

The (11) has real number solutions of Θ. However, only the presence of imaginary Θ roots ensures the oscillatory form of $u_q(t) = Ce^{\Theta t}$, according to the Euler's formula. The simplest case is when Θ is a purely imaginary number, i.e., $\Theta = j\omega$ and $\omega = 2\pi f_Q$, which leads to (12)

$$\left(\frac{RC\,\Theta + 1}{g}\right)^2 = \cos(-2\omega\tau) + i\sin(-2\omega\tau). \tag{12}$$

Equation (12) can be separated for imaginary and real parts where both must be satisfied. This leads to (13)

$$2\sqrt{1 - g^2\cos(-2\omega\tau)} = g^2\sin(-2\omega\tau). \tag{13}$$

In general (13) is satisfied for $g > 1$, $\tau > RC$ and leads to the range of $f_Q \in (1/4\tau, 1/2\tau)$ frequencies [17]. The (10)–(13) explain the oscillatory behavior of TDL in the metastable mode and were used for calculation of the inverter model estimates (in particular g from the previous subsection). However, these equations omit the nonlinear properties of the transfer functions and the presence of the real part of Θ. This makes them useless for the resolve time derivation. For this reason the delay differential (9) was solved iteratively in MATLAB with variable $u_q(t = t_{\text{rosc}})$ and $u_{\bar{q}}(t = t_{\text{rosc}})$ initial conditions and corresponding history vectors $[\mathbf{u_q}, \mathbf{u_{\bar{q}}}] \forall\, t \in (t_{\text{rosc}} - \tau, t_{\text{rosc}})$, resulting from the preceding r-osc mode in $t \in (0, t_{\text{rosc}})$. Thus, the two dimensional estimate (look-up-table) of $t_r(u_{q,t=t_{\text{rosc}}}, u_{\bar{q},t=t_{\text{rosc}}})$ dependency was derived and illustrated graphically in Fig. 5(a).

The numerical dependency presented in Fig. 5(a) allows obtaining numerical estimate of the $\widetilde{T}_r$ pdf, when $(U_Q, U_{\overline{Q}})$ pair is substituted by the r-osc random variables $\widetilde{U}_Q$ and $\widetilde{U}_{\overline{Q}}$, resulting from random $\widetilde{\varphi}_Q$ and $\widetilde{\varphi}_{\overline{Q}}$. This in turn, requires cumulative distribution functions (cdfs) of $\widetilde{U}_Q$ and $\widetilde{U}_{\overline{Q}}$ in the preceding r-osc mode. By applying a simplified description of the ISF reduced to a single harmonic, and Gaussian $\{\widetilde{\varphi}_Q, \widetilde{\varphi}_{\overline{Q}}\} \sim$

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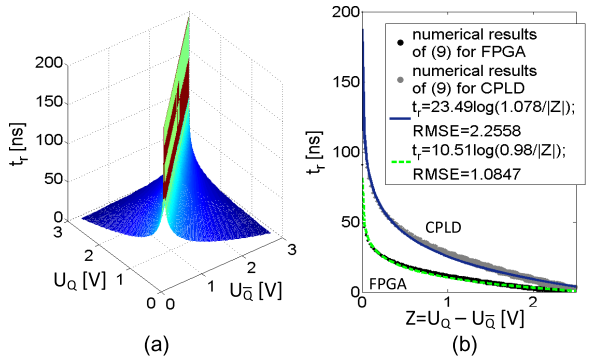

Fig. 5. Deterministic dependence of t_r vs. initial condition voltages $(U_Q, U_{\overline{Q}})$ in the oscillatory metastable operation of the TDL: (a) $t_r(U_Q, U_{\overline{Q}})$; (b) $t_r(Z)$, where $Z = U_Q - U_{\overline{Q}}$.

$\mathcal{N}(0, \sigma_\varphi(t_{\text{rosc}}))$ (see previous subsection), the cdf (denoted as $F_{\widetilde{U}_Q}(u_q)$) and the pdf (denoted as $f_{\widetilde{U}_Q}(u_q)$) of $\widetilde{U}_Q$ (and $\widetilde{U}_{\overline{Q}}$) have been derived in (14) and (15) respectively

$$\begin{aligned} F_{\widetilde{U}_Q}(u_q) &= P\{\widetilde{U}_Q < u_q\} \\ &= P\left\{A\cos(\mu_f + \widetilde{\varphi}) + U_o \le u_q\right\} \\ &= F_{\widetilde{\varphi}}\left(\cos^{-1}(y) - \mu_f\right) - F_{\widetilde{\varphi}}\left(-\cos^{-1}(y) + \mu_f\right) \\ &\quad \text{where, } \mu_f = 2\pi \mathrm{f}_Q t_{\text{rosc}},\ y = \frac{u_Q - U_o}{A}, \\ &|y| \le 1,\ A \text{ amplitude},\ U_o \text{ offset}. \end{aligned} \tag{14}$$

The symbolic differentiation of $F_{\widetilde{U}_Q}(u_q)$ combined with the substitution of $\widetilde{\varphi}$ with $\widetilde{U}_Q$ (or $\widetilde{U}_{\overline{Q}}$) leads to

$$\begin{aligned} f_{\widetilde{U}_Q}(y) &= \frac{\frac{1}{\sqrt{2\pi\sigma_\varphi^2}} \exp\left[-\frac{\left(\cos^{-1}(y)-\mu_\varphi\right)^2}{2\sigma_\varphi^2}\right]}{A\sqrt{1-y^2}} \\ &\quad + \frac{\frac{1}{\sqrt{2\pi\sigma_\varphi^2}} \exp\left[-\frac{\left(\pi-\cos^{-1}(-y)+\mu_\varphi\right)^2}{2\sigma_\varphi^2}\right]}{A\sqrt{1-y^2}} \\ &\quad \text{where,} \\ \mu_\varphi &= \begin{cases} \mu_f \bmod \pi, & \mu_f \in \langle 2l\pi, (2l+1)\pi) \\ \pi - \mu_f \bmod \pi, & \mu_f \in \langle (2l+1)\pi, (2l+2)\pi) \end{cases} \\ \mu_f &= 2\pi f_Q t_{\text{rosc}},\ |y| \le 1, l \in \mathbb{N}_0. \end{aligned} \tag{15}$$

Equation (15) is compact and defines the pdf of $\widetilde{U}_Q$ (or $\widetilde{U}_{\overline{Q}}$) for various σ_φ (defined in the previous subsection) and μ_φ, which in turn depend on t_{rosc} and the implementation (see Table I). However, according to the results shown in Fig. 5(b), t_r depends on the momentary value of $U_Q - U_{\overline{Q}}$, whereas the $t_r(U_Q - U_{\overline{Q}})$ dependency can be described with (8). Therefore, in order to derive $\widetilde{T}_r$ pdf it is necessary to numerically derive the pdf of $\widetilde{Z} = \widetilde{U}_{\overline{Q}} - \widetilde{U}_Q$ and transform it with (8), since a symbolic form of $f_{\widetilde{Z}}(Z)$ pdf does not exist. For this purpose an integral of the joint pdf of $f_{\widetilde{U}_Q}(u_q)$ and $f_{\widetilde{U}_{\overline{Q}}}(u_{\overline{q}})$ was calculated numerically by using an iterated integral (16), and differentiated. The crucial parameters of (8), i.e., $\tau_1 = 23.49$ ns, $u_1 = 1.078\,V$ and $\tau_1 = 10.51$ ns, $u_1 = 0.98\,V$ obtained for the CPLD and FPGA respectively, were calculated in MATLAB with the *lsqcurvefit* function by fitting (8) to the $t_r(U_Q - U_{\overline{Q}})$ dependency obtained iteratively with (9) [see Fig. 5(b)]

$$f_{\widetilde{Z}} = \frac{df_{\widetilde{Z}}(\widetilde{Z} < Z)}{dZ} = \frac{d\int_{U_o-A}^{A+U_o}\int_0^{u_q+z} f_{\widetilde{U}_Q} f_{\widetilde{U}_{\overline{Q}}} du_q du_{\overline{q}}}{dZ}. \tag{16}$$

In (16), bounds of the first integral result from the minimum and maximum voltage of inv1 and inv2 (forming ROs), whereas $f_{\widetilde{U}_Q}(u_q)$ and $f_{\widetilde{U}_{\overline{Q}}}(u_{\overline{q}})$ are the pdfs described with (15). The bounds of the inner integral result from a substitution of $u_{\overline{q}}$ with $u_q + z$. Calculations of (16) were done in MATLAB.

The proposed approach to finding the numerical form of $f_{\widetilde{Z}}(Z)$ allowed to identify the behavior of $\widetilde{Z}$ as a function of t_{rosc}. Moreover, the numerical description of $f_{\widetilde{Z}}(Z)$ combined with the deterministic form (8) of t_r allowed for the Monte-Carlo simulation of the $\widetilde{T_r}$ distribution. This approach leads to a variable transformation scheme (17)

$$\begin{aligned} &t_{\text{rosc}}, \{C, \tau, R, k_u\}_{\text{inv1}}, \{C, \tau, R, k_u\}_{\text{inv2}} \to \{\mu_\varphi, \sigma_\varphi\} \\ &\qquad \widetilde{\varphi}\left(\{\mu_\varphi, \sigma_\varphi\}\right) \to \{\widetilde{U}_Q, \widetilde{U}_{\overline{Q}}\} \to \widetilde{Z} \to \widetilde{T}_{rA,B}. \end{aligned} \tag{17}$$

The concept in (17) brings information on $\widetilde{T}_r$ distributions of a single TDL, whereas the target TRNG device utilizes $\widetilde{T}_r$ of two devices, i.e., $\widetilde{T}_{rA}$ and $\widetilde{T}_{rB}$, and the result of their subtraction sign, i.e., sign$(\widetilde{T}_{rA} - \widetilde{T}_{rB})$. Despite the numerical complexity resulting from the numerical integration in (17), it is much faster to estimate the whole distributions of $\widetilde{T}_{rA}$ and $\widetilde{T}_{rB}$, than to perform tedious calculations of initial condition from (1) passed to (9). Results based on (17) are presented in the next subsection.

C. Influence of r-osc Mode Length on Resolve Time Variance

The main parameter of the TDL is the t_{rosc} interval, which determines the variance of $\widetilde{Z}$ and $\widetilde{T_r}$. Assuming the lack of the TDL process variation, i.e., $\tau_Q = \tau_{\overline{Q}}$, the phase noise has an identical influence on $\widetilde{U}_Q$, $\widetilde{U}_{\overline{Q}}$, and hence on $\widetilde{T}_{rA}$ and $\widetilde{T}_{rB}$ of two different TDLs. In the case of the process variation, resulting in $\tau_Q \neq \tau_{\overline{Q}}$, the $\widetilde{U}_Q$, $\widetilde{U}_{\overline{Q}}$ pdfs reveal different variances and mean values, leading to the $\widetilde{Z}$ bias and a possible $\widetilde{T}_r$ variance loss. For this reason the analysis of a single TDL with and without process variation was performed. Therefore, the pdf transformation scheme (17) was used together with the Monte-Carlo method, while $\widetilde{Z}$ probes were generated from a numerical spline distribution of $f_{\widetilde{Z}}(Z)$ and transformed by (8). Fig. 6(a) depicts obtained $f_{\widetilde{U}_Q}(u_q)$ or $f_{\widetilde{U}_{\overline{Q}}}(u_{\overline{q}})$ distributions in two cases: i) the lack of the process variation (assumed identical parameters of inv_1 and inv_2), when the TDL switches between the r-osc and metastable mode with the maximum of ISF ($\mu_\varphi = (1/2)\pi$), and ii) when the 1.5% τ_Q and $\tau_{\overline{Q}}$ intra-class tolerance is present. The second case shifts μ_φ in $f_{\widetilde{U}_Q}(u_q)$ and $f_{\widetilde{U}_{\overline{Q}}}(u_{\overline{q}})$ distributions (denoted as $\mu_{\varphi\text{inv1}}$ and $\mu_{\varphi\text{inv2}}$ respectively), leading to the bias and skewness of $f_{\widetilde{Z}}(Z)$. The inverter frequencies

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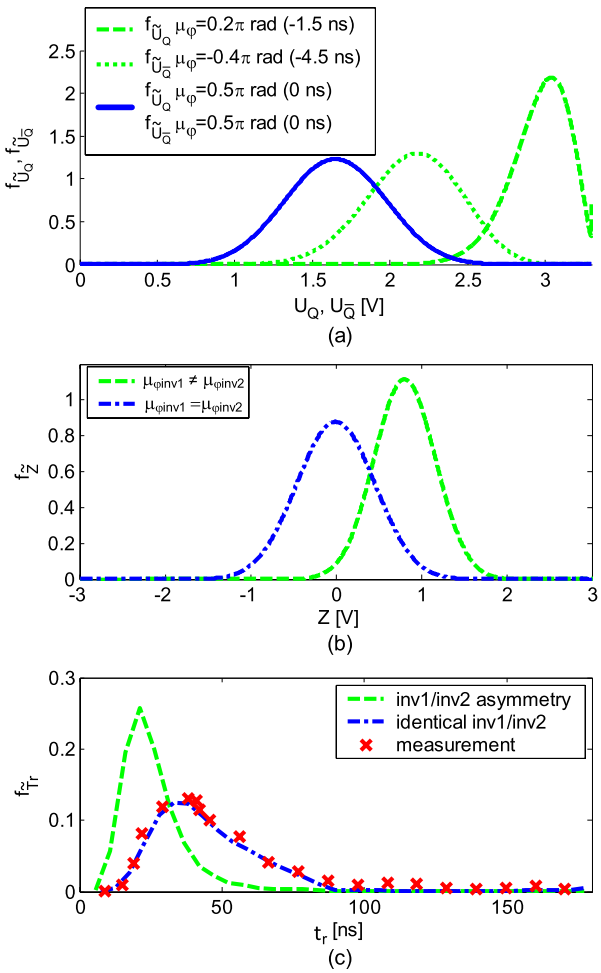

Fig. 6. Pdf transformation example with the zero and nonzero phase bias of the ring oscillators, when $t_{\mathrm{rosc}} = 200$ ns, $f_Q \approxeq f_{\overline{Q}} = 100$ MHz, and $f_Q = 99.25\ f_{\overline{Q}} = 97.8$ MHz, (a) $\widetilde{U}_Q$ and $\widetilde{U}_{\overline{Q}}$ pdfs in both cases, the relative shift from the nearest ISF maximum is shown in brackets; (b) $\widetilde{Z}$ pdfs resulting from the corresponding $\widetilde{U}_Q$ and $\widetilde{U}_{\overline{Q}}$ pdfs; (c) $\widetilde{T}_r$ pdfs obtained with the Monte-Carlo simulation based on the corresponding $\widetilde{Z}$ pdfs. The measurements for the CPLD device are also shown.

in simulations were close to 100 MHz, whereas $t_{\mathrm{rosc}} = 200$ ns. The sensitivity of t_r to Z is highest when $Z \to 0$. Therefore, when $\tau_Q \neq \tau_{\overline{Q}}$ and different μ_φ biases are present, the nonzero mean value of $\widetilde{Z}$ can decrease the $\widetilde{T}_r$ variance. This explains the behavior of the circuit in case ii), i.e., when $\mu_{\varphi\mathrm{inv1}} \neq \mu_{\varphi\mathrm{inv2}}$ presented in Fig. 6(b) and (c). One can see in Fig. 6(c) that the highest variance of $\widetilde{T}_r$ is obtained when $Z \to 0$ and $\mu_{\varphi\mathrm{inv1}} \approx \mu_{\varphi\mathrm{inv2}}$. Therefore, in the case of identical μ_φ phase biases of both $f_{\widetilde{U}_Q}(u_q)$ and $f_{\widetilde{U}_{\overline{Q}}}(u_{\overline{q}})$ pdfs, the randomness harvesting based on $\widetilde{T}_r$ would be most efficient. This particular case occurs when $f_Q = f_{\overline{Q}}$ for $t_{\mathrm{rosc}} > 0$, or when $t_{\mathrm{rosc}} = k/2|f_Q - f_{\overline{Q}}|$ for $f_Q \neq f_{\overline{Q}}$, where $k \in \mathbb{N}_1$. The second case is more interesting since it corresponds to the process variation. In order to verify the results obtained with (17), the measurement of the $\widetilde{T}_r$ distribution (of a TDL implemented in a Cool-RunnerII device) with an Agilent 53230A instrument was performed. The measured distribution in Fig. 6(c) is close to the theoretical case when $f_Q = f_{\overline{Q}}$ due to a very small physical intra-class variation.

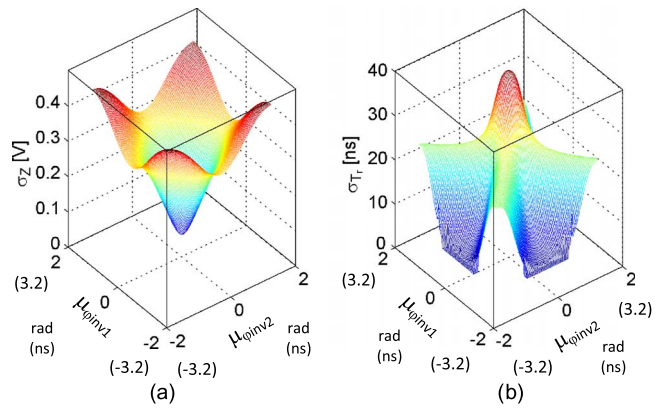

Fig. 7. Standard deviations of: (a) $\widetilde{Z}$, and (b) $\widetilde{T}_r$, as functions of $\mu_{\varphi\mathrm{inv1}}$ and $\mu_{\varphi\mathrm{inv2}}$ biases ($f_Q \approx f_{\overline{Q}} = 100$ MHz).

The results in Fig. 6 show the major effects, but neither $\mu_{\varphi\mathrm{inv1}}$, $\mu_{\varphi\mathrm{inv2}}$ influence nor the t_{rosc} optimum can be figured out. For this reason the standard deviations of $f_{\widetilde{Z}}(Z)$ and $f_{\widetilde{T}_r}(t_r)$ (denoted as σ_Z and σ_{T_r}) were numerically estimated for $\mu_{\varphi\mathrm{inv1}} \times \mu_{\varphi\mathrm{inv2}}$ pairs, within the $\pm\pi/2$ range of the ISF maximum, which correspond to all possible biases resulting from the τ or g process variation. In turn, results presented in Fig. 7(a) show that the minimum of σ_Z exists in $\{0, 0\}$, when $\Gamma(2\pi f_Q t_{\mathrm{rosc}}) = \Gamma(2\pi f_{\overline{Q}} t_{\mathrm{rosc}}) = 0$, whereas the $\max_{\mu_{\varphi\mathrm{inv1}},\mu_{\varphi\mathrm{inv2}}} \sigma_Z$ exists in $\{\pm(\pi/2), \pm(\pi/2)\}$ and its period on both axis is π. The axis in Fig. 7 are reduced to one period to increase the plot readability.

Fig. 7(b) shows the $\sigma_{T_r}(\mu_{\varphi\mathrm{inv1}}, \mu_{\varphi\mathrm{inv2}})$ dependency. Values of σ_{T_r} between moderate to maximum are observed on a $\mu_{\varphi\mathrm{inv1}} = \mu_{\varphi\mathrm{inv2}}$ or $\mu_{\varphi\mathrm{inv1}} = -\mu_{\varphi\mathrm{inv2}}$ line. The local maximum of σ_{T_r} exists in $\{0, 0\}$ and is periodic in both axis, whereas the period is π. The position of $\max_{\mu_{\varphi\mathrm{inv1}},\mu_{\varphi\mathrm{inv2}}} \sigma_{T_r}$ in $\{0 \pm n\pi, 0 \pm m\pi\}$ (where $n, m \in \mathbb{Z}$), requires an additional explanation. When σ_Z reaches its minimum in $\{0 \pm n\pi, 0 \pm m\pi\}$ (see Fig. 7(a)), the probability of reaching the $Z = 0$ neighborhood is highest due to low dispersion of $\widetilde{Z}$ values. This, in fact, causes the highest sensitivity of t_r to Z in (8) (Fig. 5(b)). Therefore, the results show that inv_1 and inv_2 should end the r-osc mode in the in-phase (or anti-phase) operation, whereas the highest σ_{T_r} is obtained when the r-osc mode stops in the ISFs minimum. The results presented in Fig. 7 can be easily expressed in terms of t_{rosc} and f_Q, $f_{\overline{Q}}$ frequency tolerance, since $\mu_{\varphi\mathrm{inv1}}$ and $\mu_{\varphi\mathrm{inv2}}$ depend on them according to (15).

In order to find $\sigma_{T_r}(t_{\mathrm{rosc}})$ dependence, a parametric Monte-Carlo simulation utilizing (17) with a variable t_{rosc} was performed in two cases: i) the in-phase operation of inv_1 and inv_2 in the r-osc mode, when $f_Q = f_{\overline{Q}}$; ii) the variable phase operation resulting from the tolerance of f_Q and $f_{\overline{Q}}$, which corresponds to τ_Q, $\tau_{\overline{Q}}$ or g process variation [16]. In the second case, the intra-class tolerance was increased to 10% and this way major effects could be observed, for a relatively small t_{rosc} in one timescale. The noise and inverter parameters were assumed for Cool-RunnerII device (see Section II-A).

Fig. 8(a) shows the mean phase difference of inv_1 and inv_2 oscillators, as a function of t_{rosc} in both considered cases.

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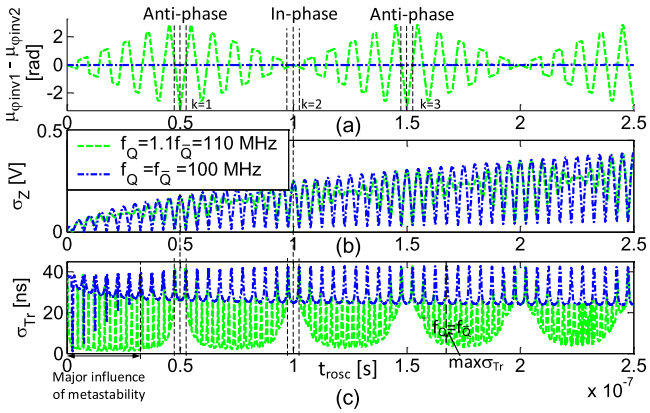

Fig. 8. Functions of t_{rosc}: (a) the mean phase difference (bias) when $f_{\overline{Q}} \neq f_Q$, and $f_{\overline{Q}} = f_Q$; (b) corresponding $\widetilde{Z}$ standard deviation; and (c) $\widetilde{T_r}$.

Figs. 8(b) and (c) show the behavior of σ_Z and σ_{T_r} respectively. In the first case (i.e., $f_Q = f_{\overline{Q}}$), the local σ_Z maximum occurs every half of a inv$_1$ and inv$_2$ period, i.e., $\max_{\mu_{\varphi\text{inv1}},\mu_{\varphi\text{inv2}}} \sigma_Z = \{k\pi, k\pi\}$, whereas $\max_{\mu_{\varphi\text{inv1}},\mu_{\varphi\text{inv2}}} \sigma_{T_r} = \{(\pi/2) + k\pi, (\pi/2) + k\pi\}$, where $k \in \mathbb{N}_1$. However, each subsequent σ_Z local maximum is greater due to the increase in $\sigma_\varphi(t_{\text{rosc}})$. In the second case (i.e., when $f_Q \neq f_{\overline{Q}}$ due to $\tau_Q \neq \tau_{\overline{Q}}$), $\sigma_{T_r}(t_{\text{rosc}})$ reaches periodic local maximums when t_{rosc} satisfies (18)

$$-\frac{1}{2(f_Q + f_{\overline{Q}})} \leq t_{\text{rosc}} - k\frac{1}{2|f_Q - f_{\overline{Q}}|} \leq \frac{1}{2(f_Q + f_{\overline{Q}})}. \quad (18)$$

One can see, in Fig. 8(c), that t_{rosc} corresponding to a few initial periods, i.e., $t_{\text{rosc}} < 30$ ns, causes a significant decrease in $\min_{t_{\text{rosc}}} \sigma_{T_r}$. In such a case, the metastable mode is primarily responsible for the $\widetilde{T_r}$ variability, and therefore the TDL becomes similar to the solution presented in Varchola *et al.* [13]. In the case of a too high t_{rosc}, the unlimited increase in σ_Z decreases the probability of reaching the $\widetilde{Z} \approx 0$ neighborhood (metastable window) and reduces σ_{T_r}. It turned out, that only in the theoretical case of $f_Q = f_{\overline{Q}}$, $\sigma_{T_r}(t_{\text{rosc}})$ has a global maximum.

The presented results show that in the case of the significant phase shift (bias) caused by the process variation, it is necessary to tune t_{rosc} for a particular TDL implementation. Otherwise randomness harvesting based on $\widetilde{T_r}$ may suffer from low entropy. Nevertheless, σ_{T_r} reaches periodical maximum values when t_{rosc} satisfies (18). Moreover, any initial voltage bias, i.e., $u_q(t = 0) \neq u_{\overline{q}}(t = 0)$, at the beginning of the r-osc mode, can be reduced with proper t_{rosc} when $f_Q \neq f_{\overline{Q}}$. This property results from the periodic form of $\mu_{\varphi\text{inv1}}(t) - \mu_{\varphi\text{inv2}}(t)$ [see Fig. 8(a)]. For this reason, the final TRNG implementation described in Section III uses the self-calibration of t_{rosc}.

D. Influence of Temperature, Voltage, Process Variation, and Frequency Attack on the Entropy

Environmental conditions influence the TDL's parameters according to Table I. Therefore, the impact of PVT on the $\widetilde{T_r}$ pdf is relevant. For this reason several experiments were conducted with the TDL under different conditions. The first

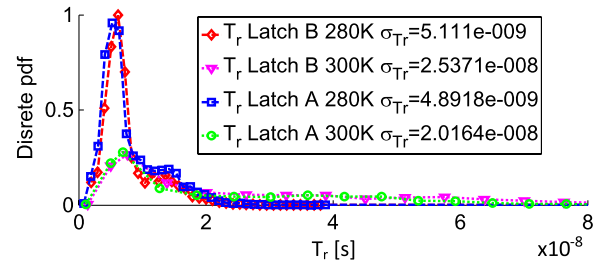

Fig. 9. Resolve time pdfs of two identical TDLs at different temperatures.

was based on the $\widetilde{T_r}$ distribution measurement of two TDLs (A and B) implemented in an FPGA Spartan 6 (Nexys 3). The $t_{\text{rosc}} \approx 80$ ns interval was manually adjusted to obtain maximum σ_{T_r} in both TDLs in 27 °C. Afterwards, the temperature was decreased to 7 °C with the Peltier thermoelectric cooler. The Fig. 9 shows the obtained $\widetilde{T_r}$ pdfs.

The $\widetilde{T_{rA}}$ and $\widetilde{T_{rB}}$ standard deviations are impressive (> 20 ns), especially at the temperature to which the t_{rosc} was adjusted. The temperature drop caused a significant decrease in variance. However, according to Table I it did not result from the σ_N loss but from the $\mu_{\varphi\text{inv1}}$ and $\mu_{\varphi\text{inv2}}$ shift, due to the f_Q and $f_{\overline{Q}}$ sensitivity to temperature. In this case $\sigma_{T_{rA}}$ and $\sigma_{T_{rB}}$ shifted from the maximum to the local minimum exactly as shown in Fig. 7(b), therefore a t_{rosc} correction is necessary.

The second experiment was to numerically verify individual entropy of two randomness sources based on TDLs, i.e., the first based on an individual $\widetilde{T_{rA}}$ or $\widetilde{T_{rB}}$ variability; and, the second, based on unpredictability of the $\widetilde{T}_{\text{diff}} = \widetilde{T_{rA}} - \widetilde{T_{rB}}$ sign. In the first randomness source the author assumed that $\widetilde{T_{rA}}$ or $(\widetilde{T_{rB}})$ is quantified by a counter incremented by subsequent metastable oscillations. The random bit is thereby produced by the youngest bit of the counter. In the second source the random bit results from sign$(\widetilde{T}_{\text{diff}})$ and is generated by an RS Arbiter [15]. Both sources can be described in terms of the entropy, i.e., $H = -[(P_0 \log_2(P_0) + (1 - P_0) \log_2(1 - P_0))]$. In the first source the P_0 stands for the probability of occurrence of such a T_{rA} (or T_{rB}) that causes the logical "1" on the youngest bit of the $\widetilde{T_r}$ (oscillation) counter. Whereas in the second source, P_0 is the probability that $\widetilde{T_{rA}} > \widetilde{T_{rB}}$. In the first source, when $\mu_{\varphi\text{inv1}}(t_{\text{rosc}}) - \mu_{\varphi\text{inv2}}(t_{\text{rosc}}) \neq 0$ then σ_{T_r} drops [see Fig. 8(c)] and reduces the number of possible metastable oscillations, which in turn affects P_0. In the second source, the bias shifts the mean values of $\widetilde{T_{rA}}$ and $\widetilde{T_{rB}}$. This in turn, biases $\widetilde{T}_{\text{diff}}$ and moves P_0 away from 0.5 [3].

Fig. 10(a) and (b) show the results of numerical H calculations for both the Cool-RunnerII and Spartan 6 devices with the parameters from previous subsections. During the entropy calculations three main cases were considered: i) t_{rosc} was initially adjusted to the maximum variance of $\widetilde{T_{rA}}$ and $\widetilde{T_{rB}}$; ii) t_{rosc} was initially adjusted to the minimum variance of $\widetilde{T_{rA}}$ and $\widetilde{T_{rB}}$, whereas afterwards in both cases τ_Q was affected by the process variation (imbalance); iii) t_{rosc} was initially adjusted to the minimum variance of $\widetilde{T_{rA,B}}$, whereas all TDL delays changed proportionally. The third case corresponds to either a temperature fluctuation, or a power supply change, whereas the first and second to improper t_{rosc} settings in the case of a delay imbalance. In the simulations the r-osc

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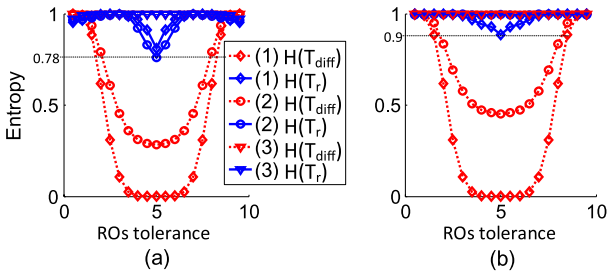

Fig. 10. Entropy in (a) CPLD and (b) FPGA obtained from metastable oscillation counting (H_{T_r}) and the $\widetilde{T}_{\rm diff}$ sign.

interval for CPLD was $t_{\rm rosc} = 50$ ns and $t_{\rm rosc} = 8.6$ ns for the FPGA, which correspond to five periods in the r-osc mode. The results in Fig. 10 show individual entropy of sources based on $\widetilde{T}_{rA,B}$ and $\widetilde{T}_{\rm diff}$ as a function of relative τ_Q and $\tau_{\overline{Q}}$ difference (R-osc tolerance). It turned out that $\widetilde{T}_{\rm diff}$ harvesting method for initially tuned $t_{\rm rosc}$ (the third case) is quite immune to changes of the temperature or voltage. The lowest $H(\widetilde{T}_{\rm diff})$ occurs when one of TDLs has the maximum σ_{T_r}, whereas the second one operates near the σ_{T_r} minimum. According to Fig. 10 the $H(\widetilde{T}_r)$ entropy is less susceptible to the process variation than $H(\widetilde{T}_{\rm diff})$. Nevertheless, the joint entropy of $H(\widetilde{T}_{\rm diff})$ and $H(\widetilde{T}_r)$ does not drop below 0.78 and 0.9 for CPLD and FPGA devices respectively.

Power supply can either influence the parameters in (2) and (9), or affect $N(t)$ by the presence of $u_{\rm det}$ component in the model presented in Fig. 2. The $u_{\rm det}$ component sums up with $N_1(t)$ and $N_2(t)$, therefore the resultant random phase walks of $\widetilde{\varphi}_Q(t)$ and $\widetilde{\varphi}_{\overline{Q}}(t)$ might lead to a partial correlation of $\widetilde{U}_Q$ and $\widetilde{U}_{\overline{Q}}$. However, according to previous results [i.e., in Fig. 5(b)], the TDL is sensitive to the $\widetilde{Z} = \widetilde{U}_Q - \widetilde{U}_{\overline{Q}}$ difference, which minimizes the presence of a common (dependent) $\widetilde{U}_Q$ and $\widetilde{U}_{\overline{Q}}$ component. A more important problem might result from an energy injection of $u_{\rm det}$ to (1), which might lead to the ROs lock-up [8]. Fig. 11(a) depicts normal TDL behavior in a Cool-RunnerII device with a significant $\widetilde{U}_Q$ and $\widetilde{U}_{\overline{Q}}$ noise when the r-osc mode ends ($t_{\rm rosc} \approx 700$ ns). However, when the frequency injection attack is present, σ_Z is significantly reduced [see Fig. 11(b)]. In this case the periodic voltage disturbances of 100 mV amplitude and 10 MHz were added to the 1.8 V core voltage of the Nexys Cool-RunnerII board. This resulted in presence of the $u_{\rm det}$ component with a $f_{u_{\rm det}}$ frequency equal to $10 f_{u_{\rm det}} \approx f_Q \approx f_{\overline{Q}}$ and synchronization of ROs within the TDL. The frequency attack led to $\widetilde{Z} \to 0$, and hence reduced the $\mu_{\varphi{\rm inv1}}(t_{\rm rosc}) - \mu_{\varphi{\rm inv2}}(t_{\rm rosc})$ bias. This in turn, improved the conditions under which the bistable enters metastability according to (8). Therefore, the metastable oscillations are still visible in the metastable mode in Fig. 11(b) despite the attack. This property has been deeply investigated by the iterative solution of (9) with $\widetilde{U}_Q$ and $\widetilde{U}_{\overline{Q}}$ obtained from (1), in the presence of $u_{\rm det} = B\sin(2\pi f_{u_{\rm det}})$ in the $N_1(t)$ and $N_2(t)$ noise components. The inverter interval imbalance during simulation was $\tau_Q = 1.015\tau_{\overline{Q}}$, whereas $t_{\rm rosc} = 680$ ns and B was changed from $0.1\sigma_N$ to $1\sigma_N$. This way an increasing influence of the deterministic component over $N_1(t)$ and $N_2(t)$ was simulated. The results are presented in Fig. 11(c)–(e). The pdfs in Fig. 11(c) show the variable bias of $\widetilde{\varphi}_{\rm inv1} - \widetilde{\varphi}_{\rm inv2}$ as a function of B (marked with an arrow). In the case of a negligible influence of $u_{\rm det}$ (small B), the inverter delay imbalance leads to the phase bias, which significantly reduces the standard deviation of $\widetilde{T}_r$ [see corresponding Fig. 11(d)]. This in turn, results in a nonzero mean value of $\widetilde{T}_{\rm diff}$ in Fig. 11(e). However, when B rises and $u_{\rm det}$ injects energy to TDLs in the r-osc mode, a significant $\mu_{\rm inv1} - \mu_{\rm inv2}$ shift towards zero can be observed [see Fig. 11(c)]. This results from partial synchronization of ROs in TDLs, which eventually leads to ROs behavior similar to the circuit symmetry. For this reason the increase in B reduces the $\widetilde{Z}$ bias, increases the $\widetilde{T}_r$ variance (the $\widetilde{T}_r$ pdf "expands"), and also shifts $\widetilde{T}_{\rm diff}$ average towards zero [see Fig. 11(e)]. Such a property of TDL reduces its phase noise in the r-osc mode during the attack, but fortunately enhances the metastable behavior and the resultant $\widetilde{T}_r$ variance.

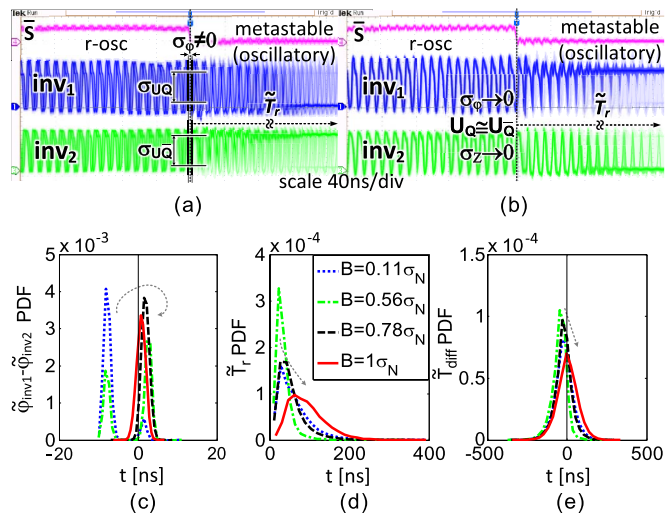

Fig. 11. Influence of frequency injection attack; (a) normal TDL operation; (b) operation during an attack; simulated TDL phase (ROs) difference pdf (c); $\widetilde{T}_r$ pdf (d); and $\widetilde{T}_{\rm diff}$ (e) behavior for variable $u_{\rm det}$.

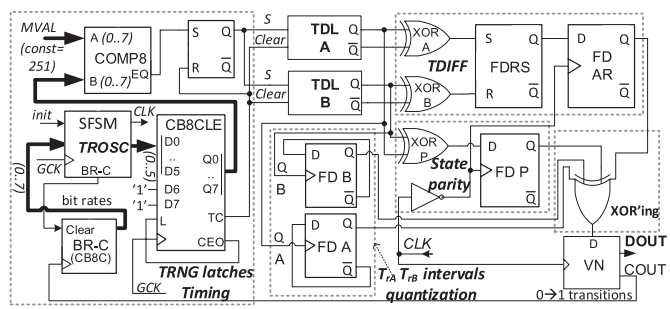

Fig. 12. Complete architecture of the proposed TRNG.

III. TRNG Implementation and Testing

The TRNG consisting of two TDL structures proposed in Fig. 1(b) was implemented both in the Cool-RunnerII XC2C265 and Spartan 6 devices. The details of the solution are presented in Fig. 12. The TRNG periodically forces the subsequent clear, r-osc, and metastable modes of both TDLs.

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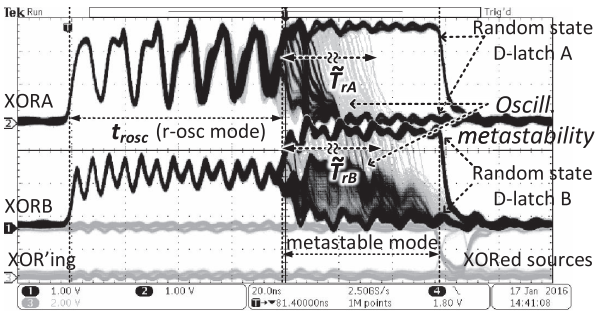

Fig. 13. Waveforms obtained with the XC6LX16 FPGA, on the XORA, XORB, XOR'ing outputs, showing the contribution of sources.

TABLE II
NIST AND DIEHARD TESTS OF THE TRNG BIT-STREAMS OBTAINED WITH DIFFERENT FPGAS AND CPLDS. SECOND-LEVEL KS TESTS WERE APPLIED TO 54 DIEHARD P-VALUES SETS. IN ALL TESTS $\alpha = 0.01$

NIST TEST	prop.	p-value
Approx. entropy	0.9875	0.3371
Block freq.	0.9906	0.9343
Cumulative sums (all)	0.9885	0.5680/0.7646
FFT	0.9917	0.6154
Frequency	0.9865	0.7061
Linear complex.	0.9917	0.5852
Longest run	0.9885	0.4296
Nonoverl. (all)	0.9902	0.5830
Overlapping	0.9927	0.6197
Rand. exc. (all)	0.9898	0.7268
Rand. exc. var. (all)	0.9908	0.3109
Rank	0.9927	0.4726
Runs	0.9917	0.1347
Serial (all)	0.9906	0.1412/0.7523
Universal	0.9938	0.5510
DIEHARD TEST SUITE	prop.	KS test p-value
The Birthday Spacings	1	0.4835
The Overl. 5-perm. (all)	0.9629	0.0414
The Binary Rank 31x31	0.9815	0.0814
The Binary Rank 32x32	1	0.0676
The Binary Rank 6x8	1	0.7626
Bit stream (all)	0.9880	0.7393
OPSO (all)	0.9760	0.3369
OQSO (all)	0.9831	0.6091
DNA (all)	0.9802	0.6094
Count 1s (all)	0.9629	0.0850
Count 1s Spec. Bytes	0.9785	0.0834
A Parking Lot	0.9629	0.4020
The Minimum Distance	0.9629	0.3194
Squeeze test	0.9444	0.7986
The 3D Spheres	1	0.5226
The Overlapping Sums	0.9629	0.3845
The Runs Test Up/Down	0.9815/1	0.6641/0.4112
The Craps Test Wins	1	0.1310
The Craps Test Throws	1	0.2988

In the solution a loadable counter (CB8CLE) is responsible for the timing of TDLs A and B. An 8 bit TROSC value passed to the CB8CLE input, determines the $(t_{\rm rosc})$ interval between subsequent S pulses of TDLs. The length of TDLs' S pulses is constant and determines the length of the metastable mode, which is equal to $(2^8 - \text{MVAL} - 1)/f_{\rm GCK}$, and can be adjusted to a particular CPLD or FPGA implementation, and the GCK frequency $(f_{\rm GCK})$. The length of the r-osc mode $(t_{\rm rosc})$ of TDLs can be easily adjusted, therefore, it is possible to increase the variance and decrease the bias of the inv_1 and inv_1 phases and $\{U_{\overline{Q}}, U_Q\}$ initial condition in both TDLs, when $S = 0 \rightarrow 1$. The metastable mode of TDLs begins when COMP8 sends logical "1" $(S = 1)$ a few cycles before the CB8CLE overflow, and ends when the CB8CLE counter sets logical "1" on the TC (terminal count) output. The optimum $t_{\rm rosc}$ value can vary and depends on many factors, as shown in previous section. For this reason the proposed solution contains an additional start-up finite state machine (SFSM), which iteratively estimates the bit rate of the TRNG for various $t_{\rm rosc}$. An 8-bit counter (BR-C) counts the positive transitions on the clock output (*COUT*) of the TRNG, for every TROSC value set iteratively by the SFSM during the start-up. Once the SFSM has checked BR-C values (*bit rates*) for all possible TROSC values (2^6), the SFSM recalls TROSC corresponding to the maximum value of BR-C (maximum *bit rate*). The maximum assumed $t_{\rm rosc}$ time (≈600 ns) corresponds to ≈60 or ≈350 r-osc cycles in CPLD and FPGA respectively, for $f_{\rm GCK} = 100$ MHz. This simple maximization (self-calibration) mechanism has been described in [3] and is performed only during the TRNG start-up (when init $= 0 \rightarrow 1$), and takes approximately 200 ms.

TABLE III
AVERAGED SYMBOL RATES OF INDIVIDUAL SOURCES AND TOTAL TRNG BIT RATES IN BOTH IMPLEMENTATIONS

↓source f_s [Hz] →	$\widetilde{T}_{rA}$	$\widetilde{T}_{rB}$	$\widetilde{T}_{diff}$	state	DOUT [kb/s]
CPLD	196	221	314	198	160
FPGA	850	920	1550	430	950

TABLE IV
AVERAGE NIST TEST PASS-RATES OF 160 MBIT LONG STREAMS OBTAINED UNDER VARIOUS CONDITIONS IN FOUR FPGA AND FIVE CPLD DEVICES. AVERAGE CPLD DOUT BIT RATES ARE ALSO INCLUDED

NIST TEST	freq. inj.	$0.9V_{core}$	$27^oC \rightarrow 7^oC$
Approx. entropy	0.9875	0.9708	0.9813
Block freq.	1	0.9813	0.9813
Cumulative sums	0.9875	0.9740	0.9970
FFT	*0.9687*	0.9792	0.9813
Frequency	0.9875	0.9748	1
Linear complex.	0.9813	0.9854	0.9938
Longest run	0.9938	0.9979	0.9875
Nonoverl. (all)	0.9908	0.9897	0.9895
Overlapping	0.9937	0.9896	0.9813
Rand. exc. (all)	0.9904	0.9860	0.9830
Rand. exc. var. (all)	0.9845	0.9877	0.9854
Rank	0.9875	0.9918	0.9938
Runs	1	0.9875	0.9938
Serial	0.9875	0.9927	0.9969
Universal	0.9875	0.9875	0.9938
Bit rate (CPLD) [kb/s]	185	100	117

The random bit-stream that is passed to a Von Neumann (VN) extractor [24] and the BR-C counter is obtained by the XOR'ing of four randomness sources. The first source (TDIFF) results from the classification of the $\widetilde{T}_{rA}$ and $\widetilde{T}_{rB}$ values with the FDRS arbiter. In this solution, the last negative slope on XORA or XORB resulting from settling of the logical state in TDLs, determines the FDRS state and a resultant random bit corresponding to the $\widetilde{T}_{\rm diff}$ sign. Afterwards, the bit is synchronized in the FDAR flip-flop. The second and third randomness sources result from the variable number of metastable oscillations, corresponding to $\widetilde{T}_{rA}$ and $\widetilde{T}_{rB}$ intervals, during the metastable mode of TDL A and B. In this solution only single random bit corresponding to the parity of oscillation number is used (as discussed in Section II-D) by FD A and B in the counter configuration. The fourth source of randomness is based on the parity of random bits produced by TDL A and B at the end of the metastable mode. Fig. 13 presents waveforms on XORA and XORB outputs of the TRNG implemented in the Spartan 6 (FPGA) device. One can see periodic extrema of

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TABLE V
COMPARISON OF MODERN TRNGS IMPLEMENTED IN VARIOUS TECHNOLOGIES

Reference	category	Device, process	scale/blocks	Mbit/s	Power mW	p. proc./control
This work	RO-osc-metast.	Xilinx FPGA/CPLD 45/180nm	271 slices/199 cells	1/0.16	90	yes/yes
Varchola et al. [14]	osc-metast.	Actel Fusion FPGA 130nm	16 TEROs	0.25	—	yes/yes
Varchola et al. [13]	osc-metast.	Actel Fusion FPGA 130nm	216 UTEROs/3 blocks	0.9	—	no/yes
Wieczorek, Golofit [15]	nearly-metast.	XilinxCPLD 350nm/discrete	7FFs + PWM + varicaps	0.6	90	yes/yes
Deak et al. [7]	PLL+metast.	Xilinx Kintex-7 28nm	2 PLLs + 2 FFs + 2 XORs	6.25	—	yes/no
Majzoobi et al. [1]	deep-metast.	Xilinx Virtex V 65nm	2 PDLs + 128 LUTs	2	—	yes/yes
Hata, Ichikawa [4]	deep-metast.	Xilinx XC4VFX20	580 slices (256 latches)	12.5	—	no/no
Rahman et al. [10]	RO	Xilinx FPGA 45 - 130nm	342-446 slices	—	—	yes/yes
Böhl et al. [11]	RO	ASIC/FPGA	<200 gates	0.05	0.48	yes/yes

different frequencies during the r-osc mode, which result from the variable $\mu_{\varphi\mathrm{inv1}}(t_{\mathrm{rosc}}) - \mu_{\varphi\mathrm{inv2}}(t_{\mathrm{rosc}})$ bias, and hence different tolerances of inverter parameters (process variation). In the particular example, SFSM tuned t_{rosc} automatically to 80 ns (which corresponds to 8 GCK cycles and ≈46 ROs cycles).

The contribution of individual entropy sources can vary as described in Section II-D, therefore the testing of the proposed TRNG contained measurements of particular sources before XOR'ing, i.e., their symbol rates (f_s) based on mean intervals between logical "0"s and "1"s. The product of XOR'ing and VN correction was tested with the second-level NIST and Diehard tests [25]–[27], moreover additional bit rate measurements were also performed. The investigated data streams and average bit rates were obtained with five different CoolRunnerII devices and four Spartan 6 FPGAs, whereas each bit-stream was 480 Mbit long. The total streams length was 4.2 Gbit. Due to a high number of 1 Mbit probes, the χ^2 test has been applied to the NIST p-values. The Diehard suite gives a smaller number of p-values than the NIST suite, for this reason a Kolmogorov-Smirnov (KS) test was used. The averaged results for the both suites are presented in Table II. Table III lists averaged symbol rates of individual sources and the total TRNG bit rates for the CPLD and FPGA implementations. Moreover, additional average NIST pass-rates of 160 Mbit long data streams obtained under various conditions are shown in Table IV. In this case the influence of reduced core supply to 90%, presence of the supply injection attack (described in Section II-D), and the temperature drop were investigated.

According to the presented results the TRNG has favorable randomness properties and passes all the second-level tests. Such high pass-rates result from the t_{rosc} adjustment to the highest bit-rate, which is related to entropy [3]. The preliminary tests of the TRNG, without the iterative TROSC adjustment revealed problems with two second-level tests, i.e., the *Frequency* (NIST) and *Count "1"s in specific bytes* (Diehard).

Table III lists the individual symbol rates, which show that each of the sources used in the TRNG contribute to the resultant entropy. Unequal f_s of $\widetilde{T}_{rA}$ and $\widetilde{T}_{rB}$ based sources, result from TDL differences (tolerance). Experiments showed that the fourth source (*state parity*) has the smallest contribution to the final bit rate, whereas the highest f_s is observed in the $\widetilde{T}_{\mathrm{diff}}$ arbiter.

Results in Table IV show a little impact of varying conditions on statistical parameters in comparison to p-values in Table II. According to the results in Fig. 9 the temperature influences the $\widetilde{T}_{rA}$ and $\widetilde{T}_{rB}$ pdfs. This explains the bit rate loss shown in Table IV. All of listed pass-rates are acceptable for a sample sizes of 160 1 Mbit sequences according to the NIST suite. However, some relationships can be seen. In the case of the frequency injection attack a slight *FFT* proportion drop occurs (marked with an asterisk), which is close to the minimum acceptable pass-rate. This results from a partial influence of the periodic attacking signal (u_{det}) on the TDLs initial condition. TDLs sensitivity to u_{det}, in turn, impacts the bit-stream's DFT spectra. The change of the core power supply gives a similar effect to temperature fluctuations; however according to Table I, the power supply gives stronger impact on the TDL parameters. This explains the similar pass-rates in both cases with several pass-rates decreased by $1, \ldots, 2.5\%$, i.e., the *Approximate Entropy*, *Cumulative Sums*, *FFT*, and *Frequency*, when the core voltage is decreased.

In short, the presented TRNG utilizing two TDLs successfully passes the NIST and Diehard second level tests; however, an additional iterative t_{rosc} adjustment mechanism is necessary, and a simple post-processing. Nevertheless, the described TRNG has several advantages in comparison to few modern solutions presented in the literature, i.e., in the logic scale and throughput. The presented results show that the device is robust to PVT, since the $\widetilde{T}_{\mathrm{diff}}$ bias is insensitive to the simultaneous inverter parameter changes. It is deceptive to compare parameters of various TRNGs without their logic scale and principle used, therefore Table V lists generation rates, logic scales, and categories of prior TRNG designs of different authors.

IV. CONCLUSION

This paper presents the TRNG, which utilizes both the ring oscillations and metastability phenomena. The solution is scalable to different technologies because its principle of operation results from the mathematical properties of the TDL topology. The circuit was validated both numerically and in the hardware. The hardware prototypes of the TRNG harvest randomness mainly from the difference of the resolve time and parity of the oscillations number. The experiments proved that the proposed technique allows for the increase of jitter's standard deviation from less than < 0.4 ns to > 20 ns, which in turn simplifies the randomness harvesting in typical reprogrammable devices. According to the presented results, the process variation or environmental conditions mainly affect the throughput of the TRNG, while the statistical parameters are maintained. Moreover, the use of ring oscillators eliminates the need for sophisticated devices like precise programmable delays. Depending on the entropy and throughput, the TRNG self-adjusts the t_{rosc} parameter to satisfy particular implementation and environmental conditions.

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